

Skill 6

Single and Double Quoting

Aim

To implement single and double quotation handling in shell commands, preserve literal content, control variable expansion, preserve spaces and validate quoted command outputs.

Requirements

- Ubuntu Linux
- GCC
- Terminal

Step-by-Step Implementation

Part A — Single Quotes

Step 1: Enter a Single-Quoted String

Example:

```
echo 'Hello World'
```

The complete text is treated as a single literal string.

Step 2: Preserve Literal Content

Characters inside single quotes are preserved without interpretation.

For example:

```
echo '$HOME'
```

produces the literal text:

```
$HOME
```

instead of expanding the variable.

Step 3: Store Quoted Strings

The parser identifies the beginning and ending single quotes and stores the content as one token.

Part B — Double Quotes

Step 4: Enter a Double-Quoted String

```
echo "Hello World"
```

The spaces are preserved as part of the same argument.

Step 5: Allow Variable Expansion

For example:

```
echo "$HOME"
```

allows the shell to expand the HOME environment variable.

Step 6: Parse Quoted Tokens

The parser identifies quoted regions and prevents spaces inside the quotes from incorrectly separating tokens.

Step 7: Test Edge Cases

Examples:

```
echo 'hello world'  
echo "hello world"  
echo '$HOME'  
echo "$HOME"
```

Observation

Single quotes preserved their contents literally, while double quotes preserved spaces and allowed variable expansion. The parser correctly identified quoted regions and treated their contents as appropriate command arguments.

End Result

Single-quote and double-quote parsing was successfully implemented and tested with normal and edge-case inputs.

